Vipul Sapra

Charlotte, North Carolina | 513-693-6897 | vpsapra5@gmail.com | linkedin.com/in/vipul-sapra-ab30a09a

EDUCATION

University of Cincinnati, Carl H. Lindner College of Business — **Cincinnati, OH**
Master of Science in Information Systems and Graduate Certificate in Data Science — **April 2022**
Cumulative GPA: 3.95 / 4.0

Pune University, Maharashtra Institute of Technology — **Pune, India**
Bachelor of Engineering in Petrochemical — **June 2017**

TECHNICAL SKILLS

Concepts: Distributed Data Processing, Dimensional Modeling, Star/Snowflake Schemas, SCD Type 1/2, Predictive Modeling, Machine Learning Operations (MLOps), Automated Data Quality Auditing, Root Cause Analysis (RCA), CI/CD Pipelines, Cloud Infrastructure Optimization

Programming Languages & Tools: Databricks, Snowflake, AWS (S3, Redshift), Azure (ADF, Synapse, Azure SQL), Apache Spark, PySpark, Python, XGBoost, Scikit-Learn, Power BI, Tableau, Airflow, KIRO IDE

Certifications: Data Science Graduate Certificate – University of Cincinnati

PROFESSIONAL EXPERIENCE

Charter Communications — Charlotte, NC

Lead Software Engineer - Databricks & Cloud Data Platforms — **May 2022 – Present**

- **Architected high-throughput data processing pipelines using** Azure Databricks, PySpark, and Apache Spark to drive scalable financial analytics and high-performance SQL stored procedures, adhering to rigorous JPMorganChase data governance standards.
- **Designed enterprise-grade dimensional data models leveraging** Snowflake, Star/Snowflake schemas, and Slowly Changing Dimensions (SCD Type 1/2) to support complex enterprise reporting requirements and maintain 99.9% data availability SLAs.
- **Engineered proactive anomaly detection and predictive modeling systems with** XGBoost, Random Forests, and Scikit-Learn, mitigating ETL discrepancies by 95% and boosting operational reliability by 20% in AWS and Azure cloud environments.
- **Implemented automated data quality audit frameworks and firewalls utilizing** Azure Data Factory (ADF), AWS S3, and Redshift, drastically cutting operational turnaround times by over 80% and ensuring pristine data integrity.
- **Spearheaded Agentic AI and automated QA testing frameworks powered by** KIRO IDE and prompt engineering to accelerate software delivery pipelines by 40% while upholding rigorous security and compliance protocols.

Master of Science in Information Systems & Data Science Graduate — **May 2022 – Present**

- **Mastered advanced database management and analytical methodologies with** a stellar 3.95/4.0 GPA from the University of Cincinnati, focusing on information systems architecture.
- **Conducted rigorous academic research and practical implementations in** machine learning algorithms, predictive analytics, and high-performance computing frameworks.
- **Synthesized foundational software engineering paradigms through** comprehensive study of distributed data processing, system design, and data governance.

Bachelor of Engineering in Petrochemical — **May 2022 – Present**

- **Applied rigorous mathematical and analytical frameworks from** Pune University MIT to solve complex optimization and process control engineering problems.
- **Developed strong engineering fundamentals emphasizing systematic troubleshooting** root cause analysis (RCA), and structured problem-solving methodologies.
- **Collaborated on multi-disciplinary engineering projects requiring** precise execution, rigorous testing, and cross-functional technical communication.

Ameritas Insurance — Cincinnati, OH

Data Governance Intern — **September 2021 – April 2022**

- **Co-engineered the enterprise-wide Data Advocacy & Governance Program** across insurance portfolios, establishing foundational frameworks for data democratization and organizational accountability.
- **Architected a comprehensive Data Intelligence ecosystem in Collibra**, establishing a centralized Data Dictionary, automated Data Quality rules, and end-to-end **Data Lineage** to ensure regulatory compliance and transparency for financial and insurance assets.

Accenture — Pune, India

Test Engineering Analyst — **September 2017 – December 2020**

- **Orchestrated end-to-end Quality Assurance and system integration testing** for mission-critical client applications at **Allianz Insurance**, leveraging Agile SDLC methodologies to ensure rapid, zero-defect feature deployments.
- **Spearheaded the Health Insurance module QA strategy**, directing cross-functional efforts for regression testing, test data validation, and the seamless rollout of complex new business functionalities.
- **Drove operational excellence by automating core regression test suites**, significantly increasing engineering productivity and reducing manual testing overhead through scalable automation frameworks.